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(54) **MARKER PROTECTION SYSTEM**

(71) Applicant: **reMarkable AS**, Oslo (NO)

(72) Inventors: **Gaute Wiig Nordby**, Oslo (NO); **Xian Dong Guo**, Pudong (CN); **Peng Shuai**, Shenzhen (CN); **Wen Bin Zhou**, Shenzhen (CN); **Liang Xiong**, Shenzhen (CN)

(73) Assignee: **reMarkable AS**, Oslo (NO)

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See application file for complete search history.

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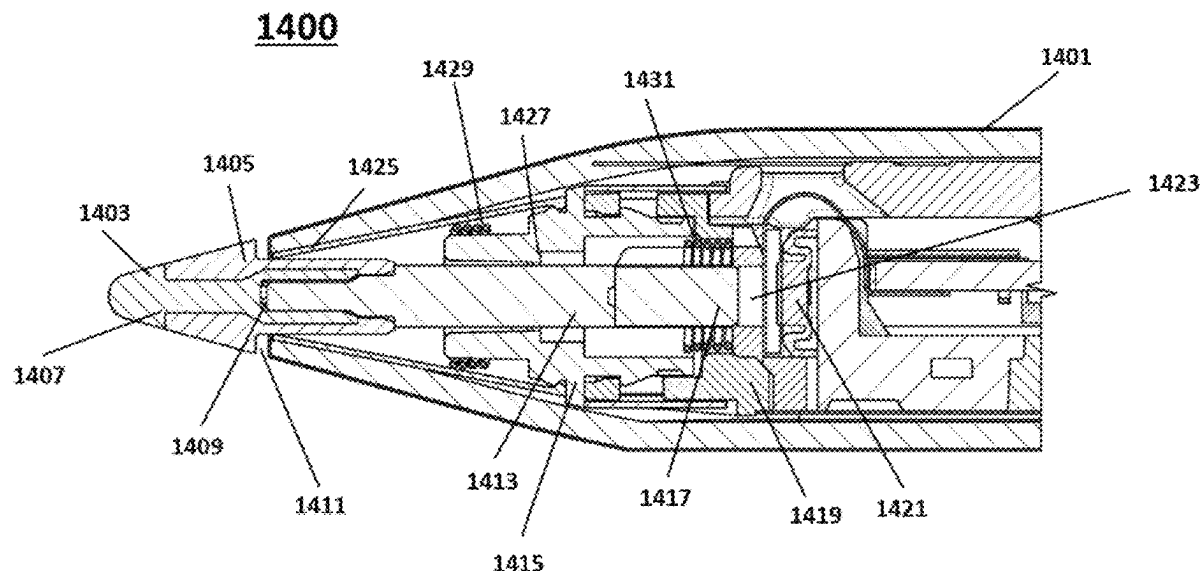
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Primary Examiner — Temesghen Ghebretinsae
Assistant Examiner — Ivelisse Martinez Quiles
(74) *Attorney, Agent, or Firm* — AWA Sweden AB;
Thomas L. Ewing

(57) **ABSTRACT**

Embodiments of the invention provide a plurality of safety features for a pen-stylus. The pen-stylus may comprise a writing shaft that receives mechanical input forces as a user engages the pen-stylus on a display of a tablet device, a force sensor that translates mechanical input forces received from the writing shaft into electrical ones for operation by electronics in the pen-stylus; and a writing shaft impact shock absorber that receives heavy forces imparted to the pen-stylus to shield the force sensor from receiving such heavy forces. The heavy forces impacting the pen-stylus may arise from the pen-stylus being dropped onto a hard surface.

20 Claims, 10 Drawing Sheets



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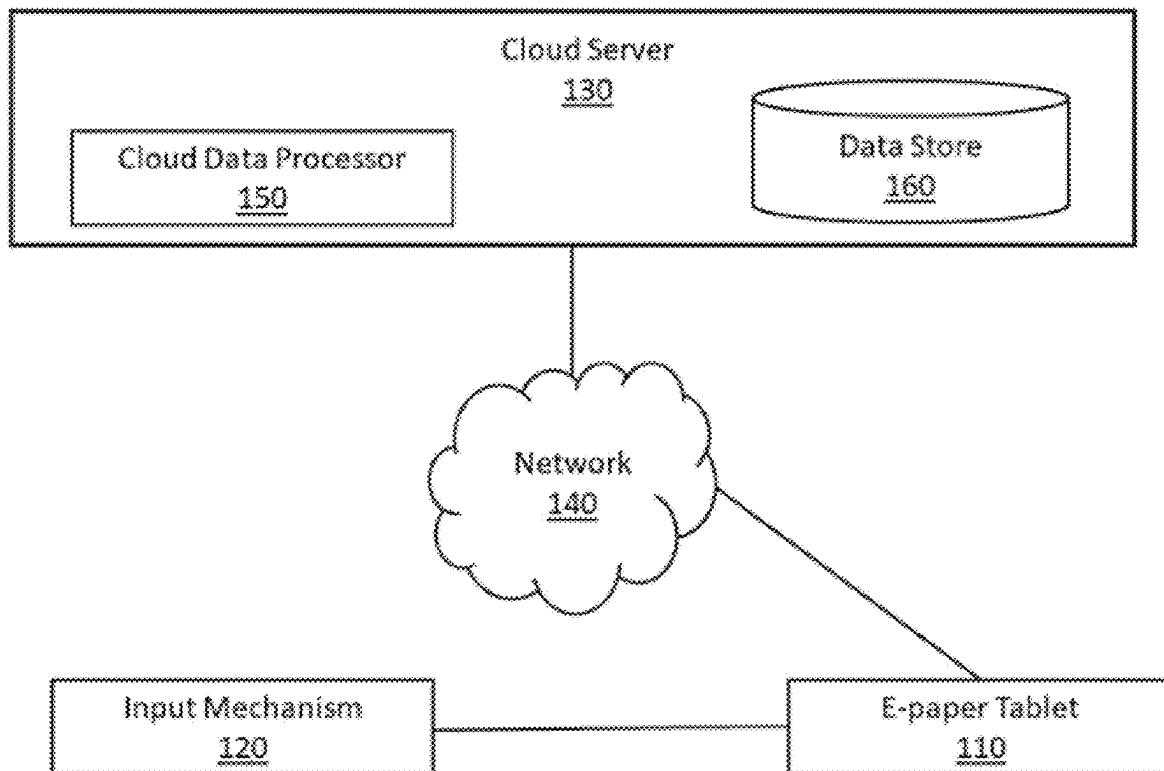
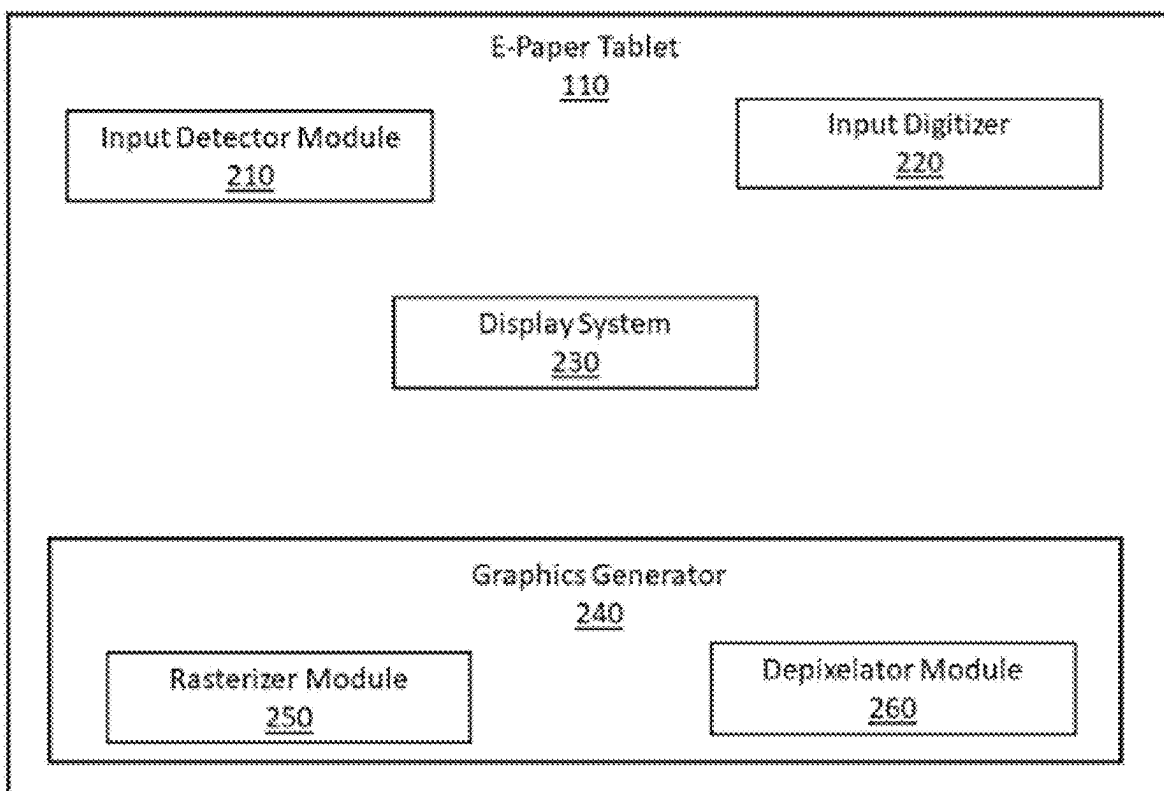
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**FIG. 1****FIG. 2**

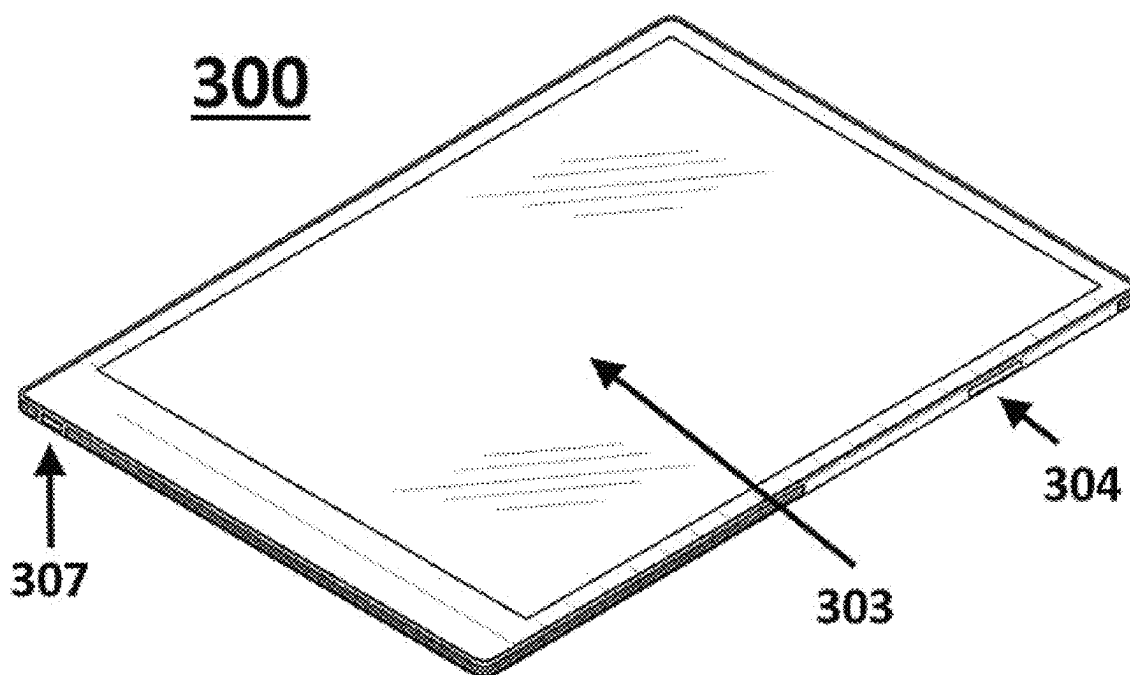


FIG. 3

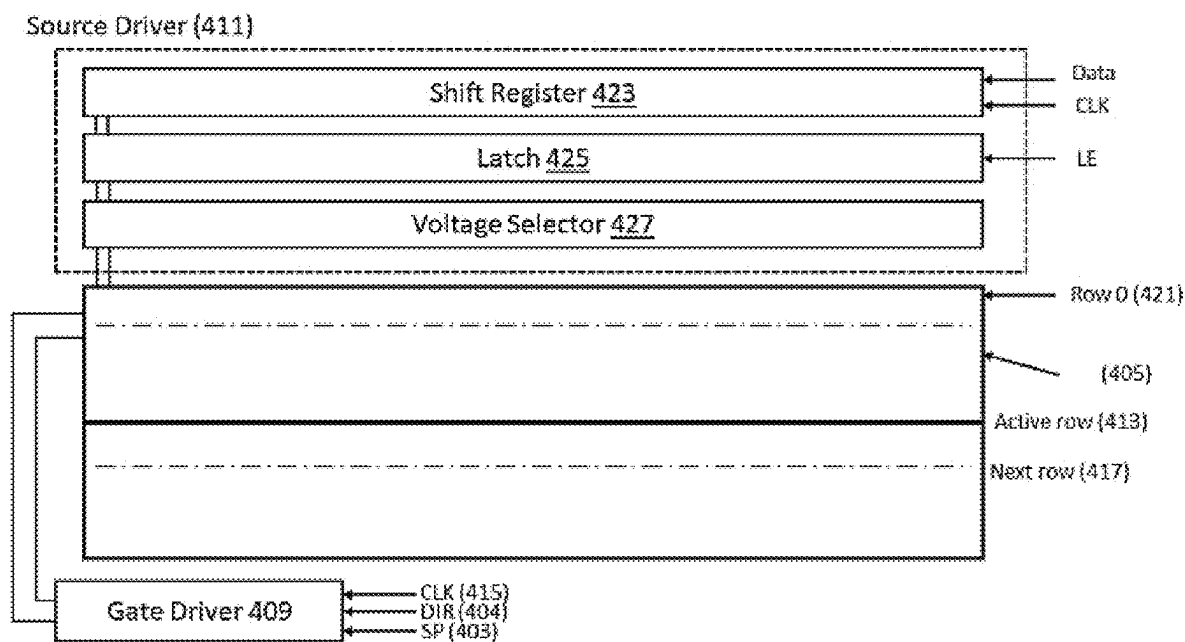
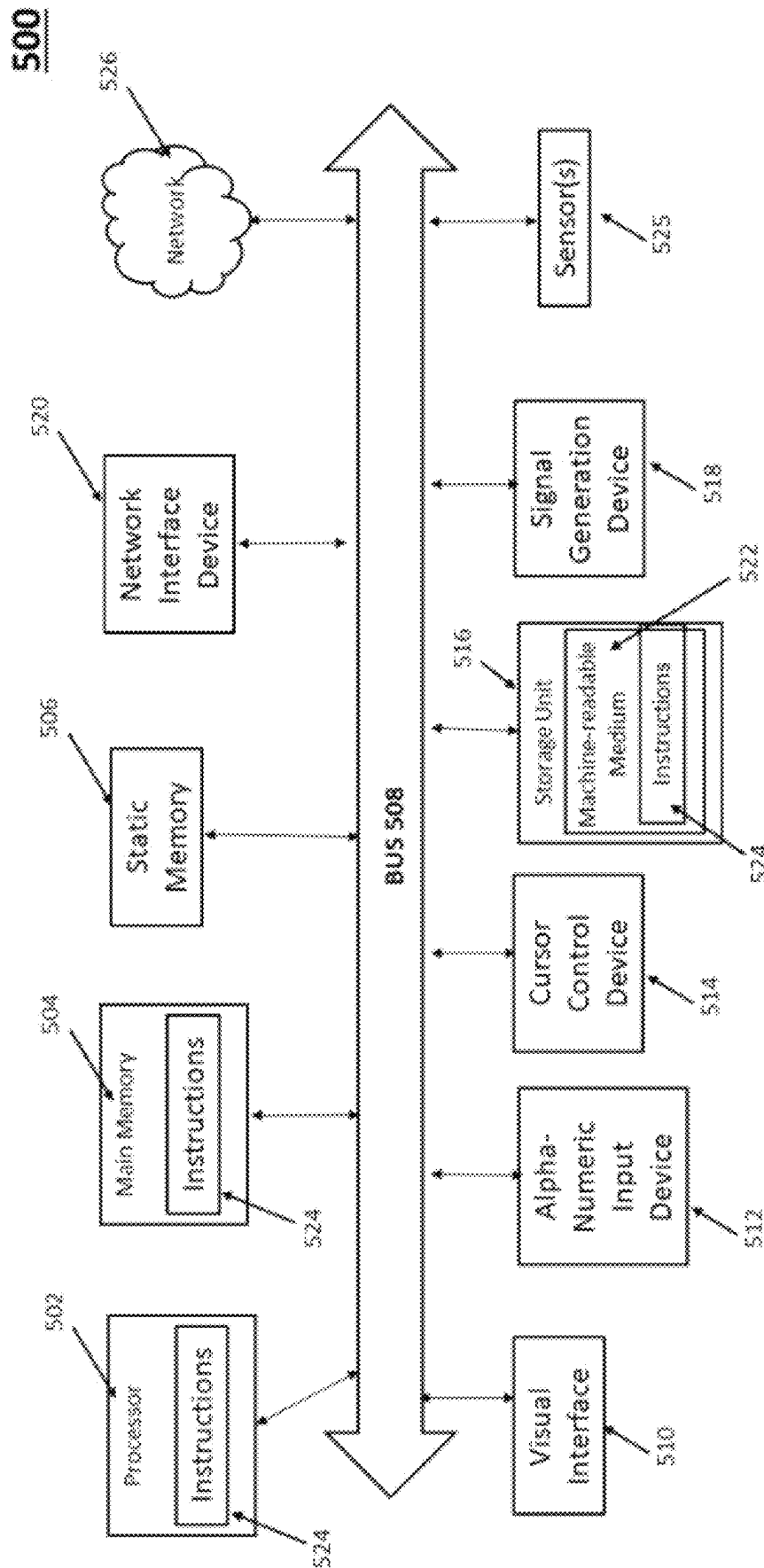


FIG. 4

**FIG. 5**

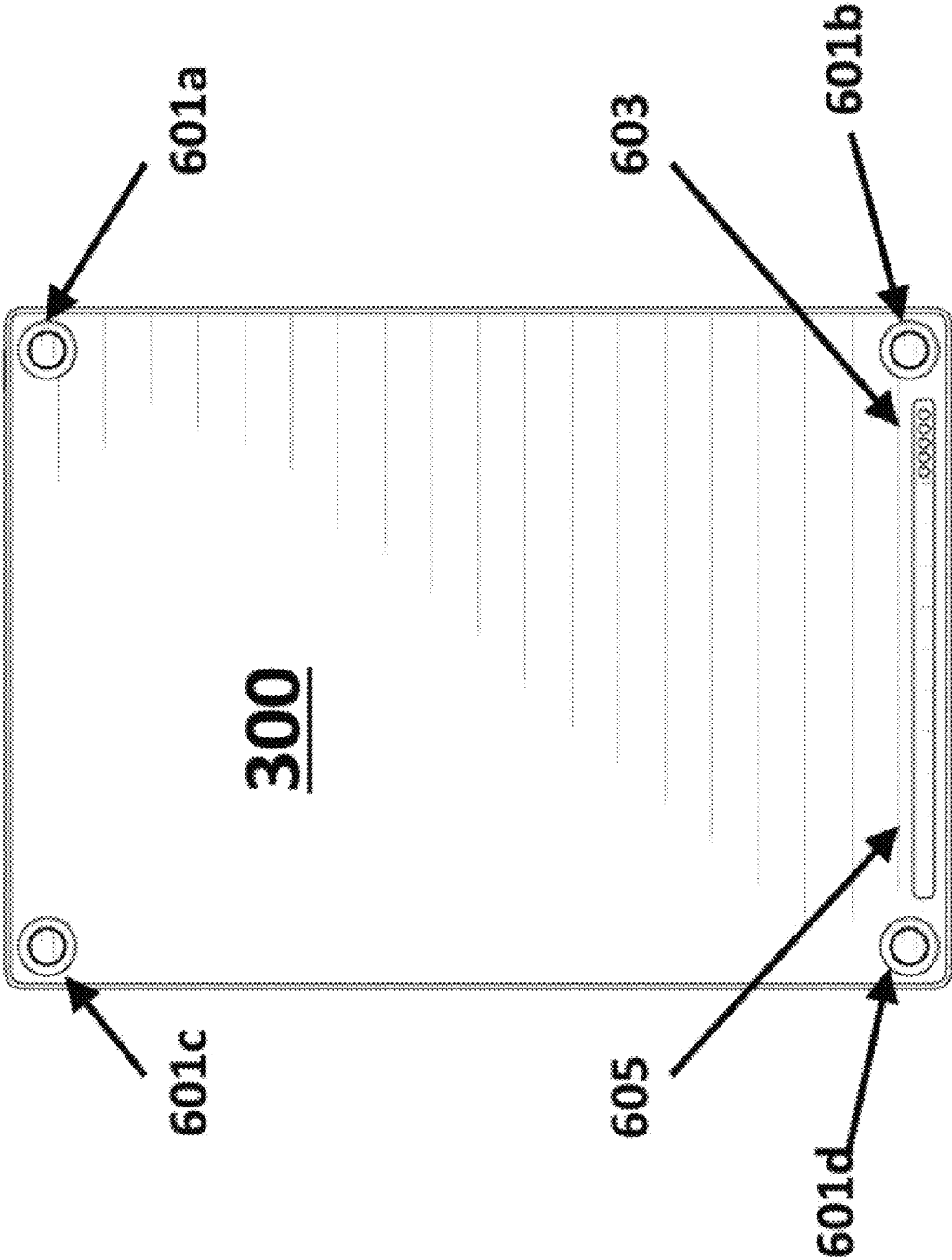


FIG. 6



FIG. 7

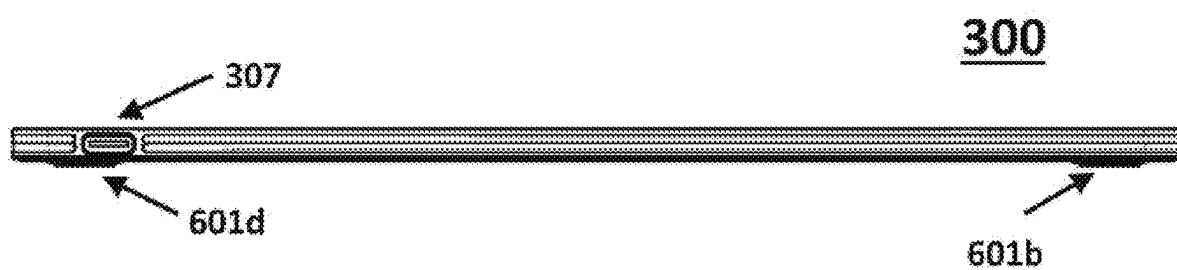


FIG. 8

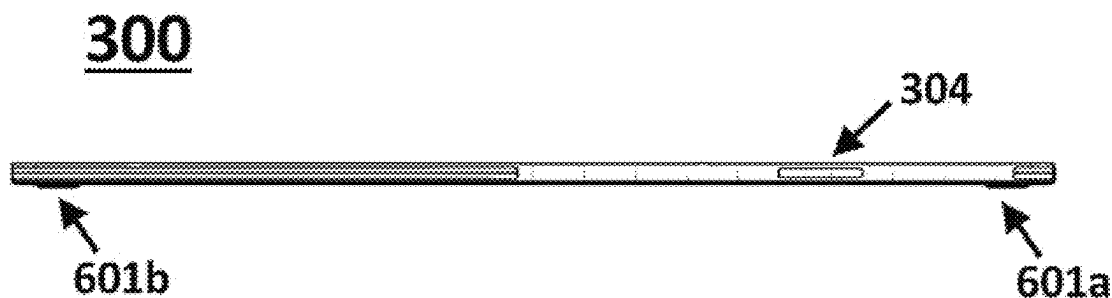


FIG. 9

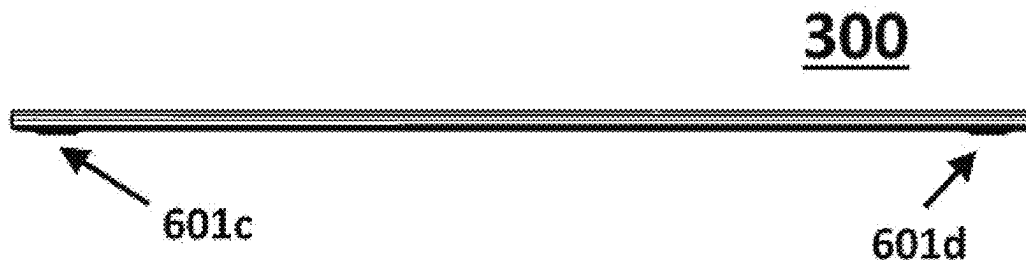


FIG. 10

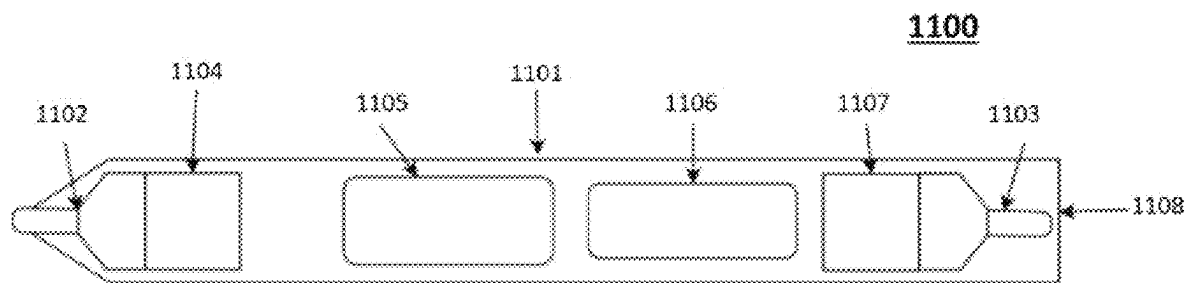


FIG. 11

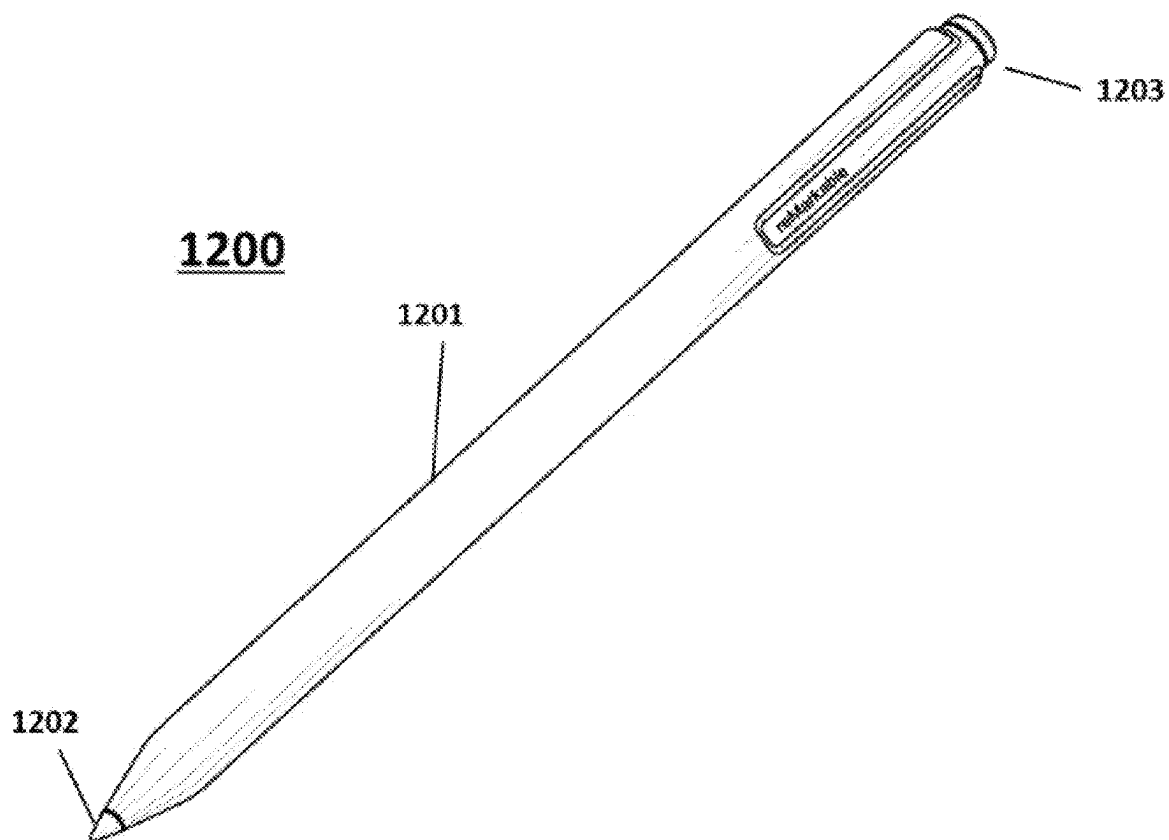


FIG. 12

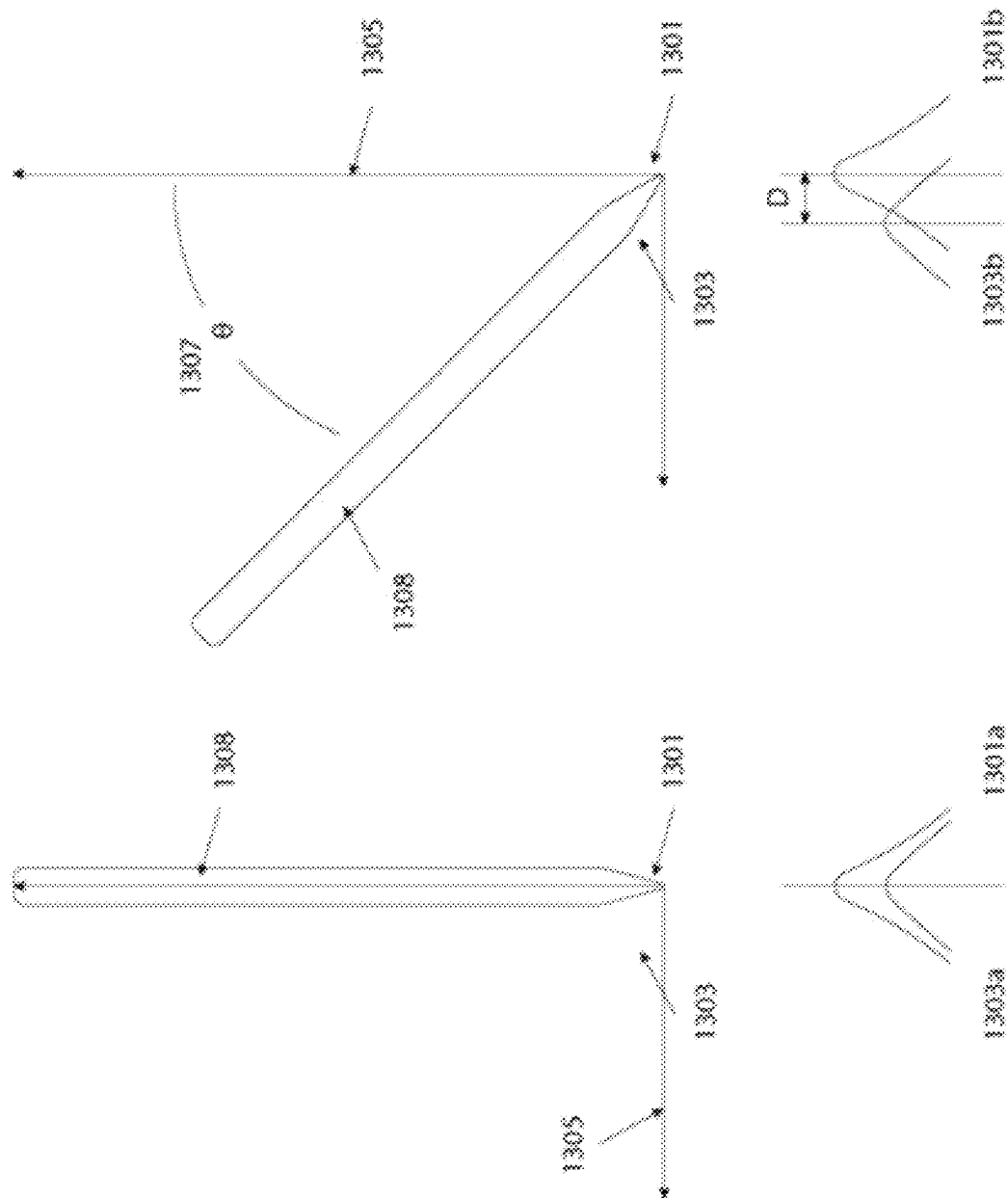


FIG. 13A

FIG. 13B

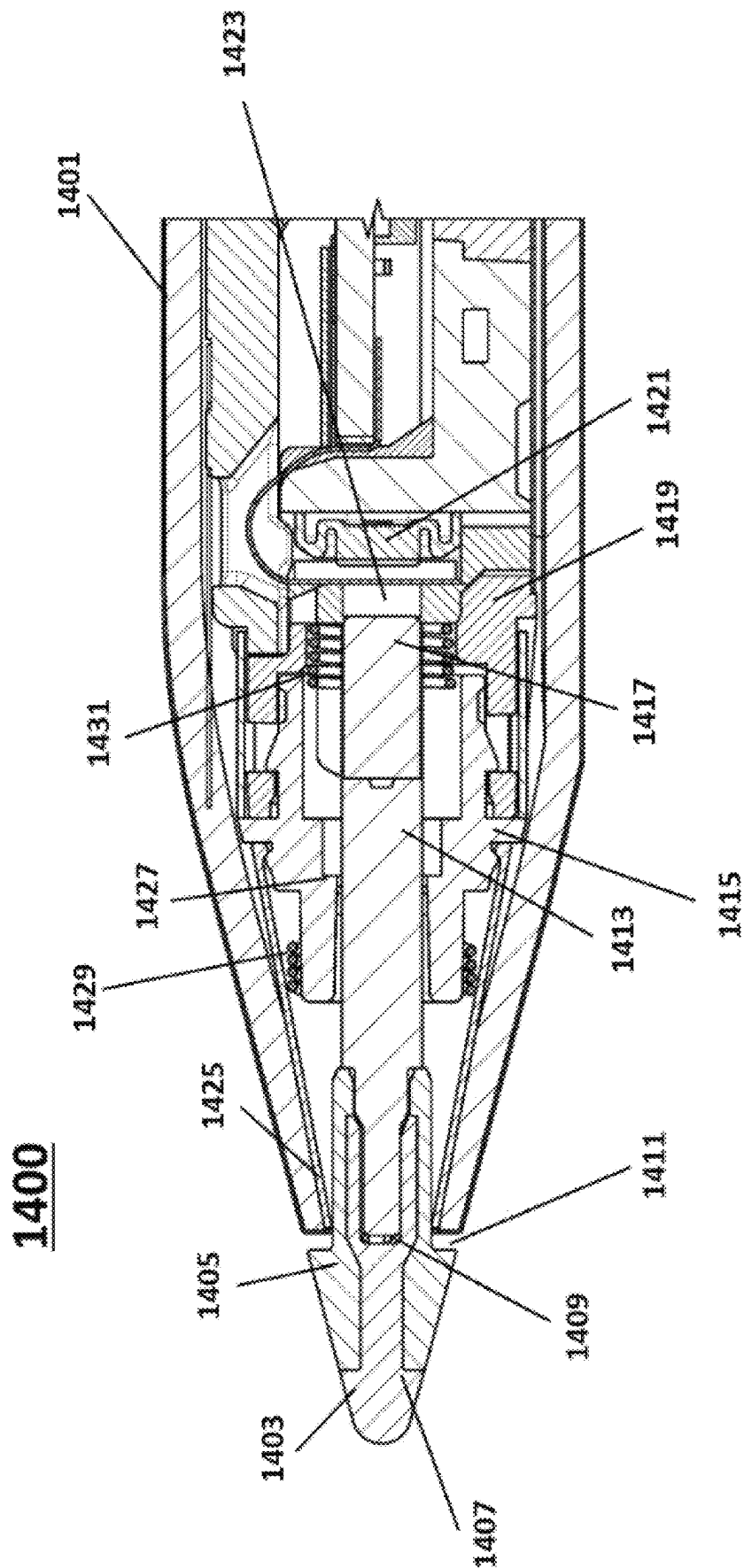


FIG. 14

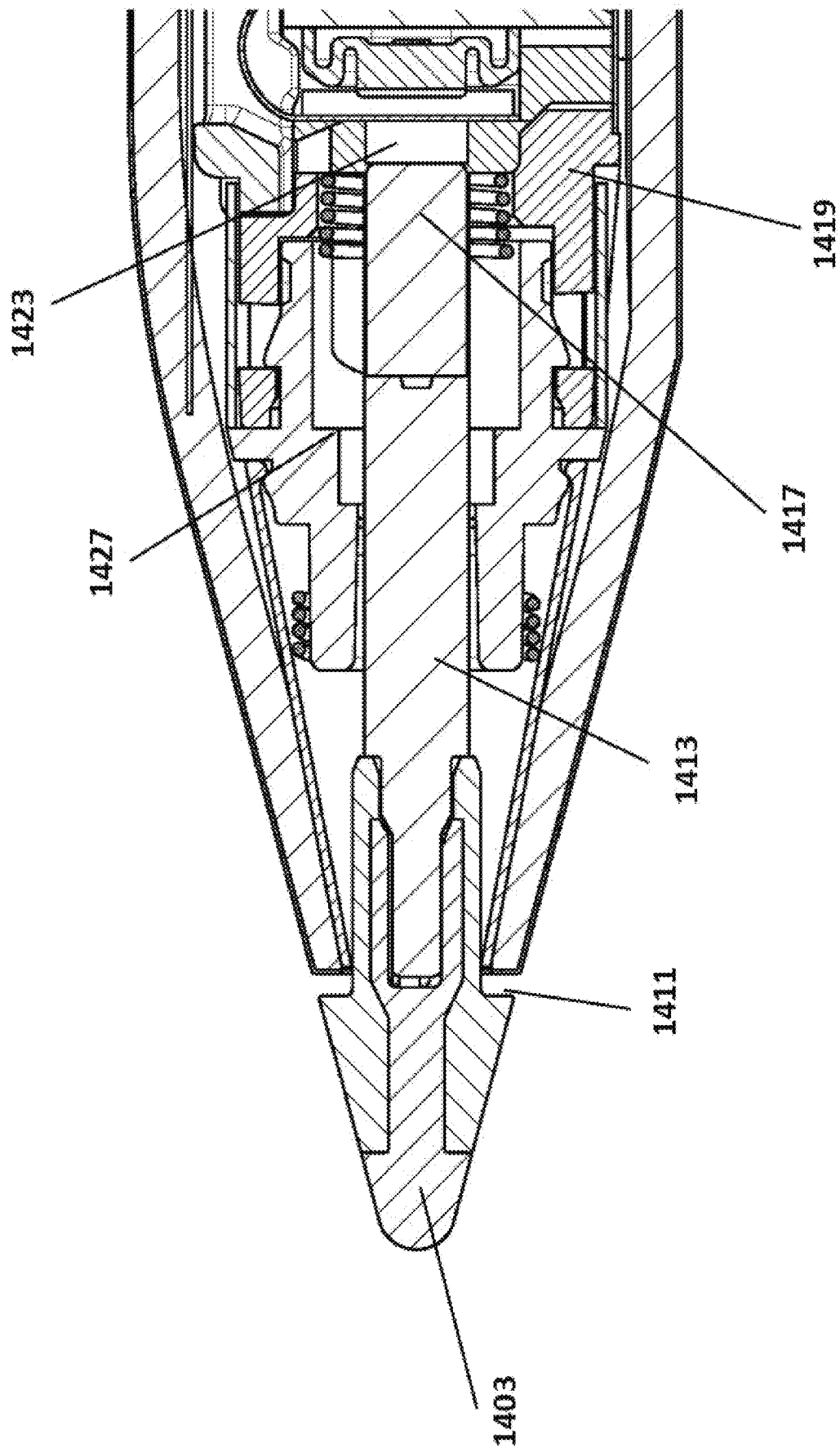


FIG. 15

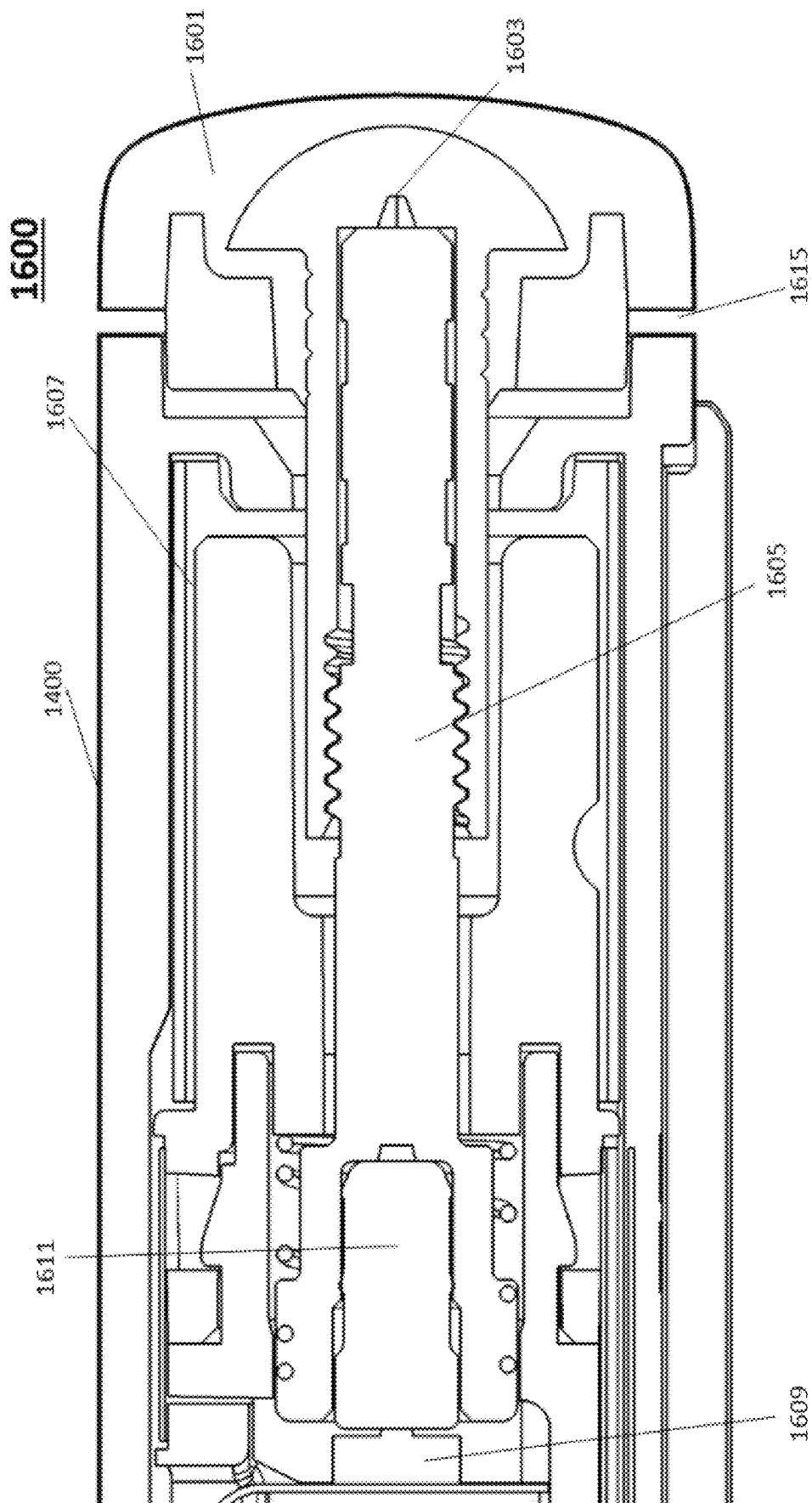


FIG. 16

MARKER PROTECTION SYSTEM**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application, U.S. patent application Ser. No. 18/779,158 entitled “Marker Protection System,” is related to U.S. patent application Ser. No. 18/779,151, entitled “Replaceable Conductive Marker Tip,” filed Jul. 22, 2024; U.S. patent application Ser. No. 18/779,154, entitled “Advanced Paper Emulation,” filed Jul. 22, 2024; U.S. patent application Ser. No. 18/779,164 entitled “Marker Writing System,” filed Jul. 22, 2024; U.S. patent application Ser. No. 18/779,170 entitled “Captive Object Flexure Mechanism,” filed Jul. 22, 2024; and U.S. patent application Ser. No. 18/779,149, entitled “Active Pen-Stylus Precise Eraser,” filed Jul. 22, 2024, all of which are owned by the Applicant of the present application. These related applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The disclosure relates generally to a pointing device, adapted for various coordinate input devices such as a digitizer or a tablet, which provide inputs to various types of computing systems. In particular, embodiments of the present invention relate to safety features that extend the useable service lifetime for a pen-stylus.

BACKGROUND

Mobile telephones, tablet computers, PCs, car entertainment systems, white goods and many other devices are commonly equipped with interactive displays. These interactive displays combine a display screen, such as an LCD, OLED, plasma or electrophoretic display (EPD), with an input system, such as a touch- or pen-stylus-input system. The input system recognizes the presence of an input object such as a pen-stylus touching or in proximity to the display screen. The device typically responds to such inputs by performing one or more functions, which may include changing what is shown on the display screen.

A “pen-stylus” (or “pen” or “stylus”) is typically a pen- or pencil-shaped instrument whose position (e.g., tip position) on a computer monitor can be detected either electronically or physically. The pen-stylus enables users to perform tasks, such as drawing or making selections on a computing device. While devices with touchscreens, such as some computers, mobile devices (smartphones and personal digital assistants), game consoles, and graphics tablets, can often be operated with a fingertip, a pen-stylus typically provides more accurate and controllable input. In essence, a pen-stylus has a similar function as a mouse or touchpad as a pointing device but may enable much more precise inputs for certain drawing tasks. The use of a pen-stylus is sometimes termed “pen-stylus computing.”

Conventional pen-styluses have typically been constructed to detect “pen-down” information in addition to coordinate information on the pointing device. Such pen-down information typically arises when the pen-stylus point is in contact with a panel of the digitizer. The pen-down information is conventionally detected by either force (e.g., pressure) sensitive means for detecting the vertical force applied to the pen-stylus point and/or detected by an electrical connection between the pen-stylus and the panel of the digitizer. The position data may be smoothed and/or de-noised before it is used to estimate the velocity and/or the

acceleration of the input object. Such smoothing and/or de-noising may be done using an appropriate technique—for example, by applying a recursive Bayesian filter or smoothing, such as a Kalman filter, to the position data.

Active pen-styluses (also known as “active pen” or “digital styluses”) include digital components and/or circuitry inside the pen-stylus that communicates with a digitizer on the touch device. This communication allows for advanced features such as force (e.g., pressure) sensitivity, tilt detection, programmable buttons, palm detection; eraser tips, memorizing settings, and writing data transmission.

Active pen-styluses typically employ different protocols from different manufacturers in order to communicate with the digitizer of a graphic tablet or multi-touch device. For an active pen-stylus to function properly, its digital component protocol must typically match the digitizer technology in the touch screen with which it interacts. Thus, the digital protocol of the pen-stylus must be compatible with the device digitizer, otherwise input from the pen-stylus will not register on the device. Active pen-styluses are typically powered by a removable or chargeable battery.

A pen-stylus’ performance is often measured by four characteristics: 1) comfort, 2) resistance, 3) balance and overall weight, and 4) precision. “Precision” can sometimes be a nebulous characteristic, so it is often described in terms of further characteristics, such as: 1) responsiveness and speed, 2) jitter, 3) tilt, 4) levels of force (e.g., pressure), and 5) palm rejection or detection. This last element of precision may prevent a touch device from registering or marking the screen when a hand or palm is resting on the screen surface. Effective operation may rely on a combination of technology in the pen-stylus, the operating system software and the screen digitizer technology for effective operation.

While pen-stylus technology has made great strides in recent years in improving pen-stylus technology, further improvements are still warranted. Moreover, specific use cases for pen-styluses may compel levels of precision and additional functionality not available in conventional devices.

SUMMARY OF THE INVENTION

Embodiments of the invention provide a plurality of safety features for a pen-stylus. The pen-stylus may comprise a writing shaft that receives mechanical input forces as a user engages the pen-stylus on a display of a tablet device, a force sensor that translates mechanical input forces received from the writing shaft into electrical ones for operation by electronics in the pen-stylus; and a writing shaft impact shock absorber that receives excess forces imparted to the pen-stylus to shield the force sensor from receiving such excess forces. The excess forces impacting the pen-stylus may arise from the pen-stylus being dropped onto a hard surface. Additional safety elements may include a marker tip that receives excess forces and deforms its shape in absorbing the excess forces and/or even shears off from the pen-stylus. Further safety features on a front end of the pen-stylus include a writing shaft safety stop and a writing shaft bottom bracket. The rear end of the pen-stylus may include further safety features such as an eraser shaft impact shock absorber, according to an embodiment of the invention.

BRIEF DESCRIPTION OF DRAWINGS

The disclosed embodiments have other advantages and features which will be more readily apparent from the

detailed description, the appended claims, and the accompanying figures (or drawings). A brief introduction of the figures is below.

FIG. 1 illustrates a system architecture for an e-paper tablet device **110** that receives inputs from the input mechanism such as a pen-stylus, according to one example embodiment.

FIG. 2 is a block diagram of the system architecture of an e-paper tablet device **110**, according to one example embodiment.

FIG. 3 illustrates a front and right perspective view of an e-paper tablet **300** having the functionality described for the e-paper tablet device **110** in FIGS. 1-2.

FIG. 4 illustrates hardware components of an example Electrophoretic Display (EPD) operating in an e-paper tablet device **110**, according to one example embodiment.

FIG. 5 is a block diagram illustrating components of an example machine able to read instructions from a machine-readable medium and execute them in a processor (or controller), according to one example embodiment.

FIG. 6 illustrates a rear view of the e-paper tablet **300** showing volcano feet **601a-601d**, a pogo pad **603**, and an antenna region **605**, according to an embodiment of the invention.

FIG. 7 illustrates a top view of the e-paper tablet device **300** showing volcano feet **601a**, **601d**, and a power button **701**, according to an embodiment of the invention.

FIG. 8 illustrates a bottom view of the e-paper tablet device **300** showing volcano feet **601b**, **601d** and the USB-c connector **307**, according to an embodiment of the invention.

FIG. 9 illustrates a right view of the e-paper tablet device **300** showing volcano feet **601b**, **601a**, and the charging area **304** for recharging the input device **120**, when the input device is an active pen-stylus, according to an embodiment of the invention.

FIG. 10 illustrates a left view of the e-paper tablet device **300** showing volcano feet **601c**, **601d**, according to an embodiment of the invention.

FIG. 11 illustrates a pen-stylus **1100** suited for application in at least one embodiment of the invention.

FIG. 12 illustrates an external casing **1201** for a pen-stylus **1200** designed to fit a user's hand, according to an embodiment of the invention.

FIGS. 13A-13B illustrate two transmitters **1301**, **1303** in a marker's tip (e.g., the core **1102** shown in FIG. 11) of an active pen-stylus **1308** giving a computing device (e.g., the e-paper tablet **110**) the ability to measure two distinct signals.

FIG. 14 illustrates a cross section of the forward portion of an active pen-stylus **1400** that a user may employ for tasks such as drawing a line on the display of a device (e.g., the e-paper tablet **110**), according to an embodiment of the invention.

FIG. 15 provides an exploded view of the forward end of the pen-stylus **1400** shown in FIG. 14 and particularly points out various device safety features found in the forward section of the pen-stylus **1400**, namely, the writing shaft impact shock absorber **1417**, the front marker safety gap **1411**, the writing shaft safety stop **1427**, and the writing shaft bottom bracket **1419**, according to an embodiment of the invention.

FIG. 16 illustrates a rear portion **1600** of the pen-stylus **1400** that includes an eraser cap **1601**, a first rear antenna **1603**, a second rear antenna **1607**, a rear erasure shaft **1605**,

a rear eraser gap **1615**, a rear impact shock absorber **1611**, and a rear force sensor **1609**, according to an embodiment of the invention.

The figures depict various embodiments of the presented invention for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

DETAILED DESCRIPTION

The Figures (FIGS.) and the following description relate to preferred embodiments by way of illustration only. It should be noted that from the following discussion, alternative embodiments of the structures and methods disclosed herein will be readily recognized as viable alternatives that may be employed without departing from the principles of what is claimed.

Reference will now be made in detail to several embodiments, examples of which are illustrated in the accompanying figures. It is noted that wherever practicable similar or like reference numbers may be used in the figures and may indicate similar or like functionality. The figures depict embodiments of the disclosed system (or method) for purposes of illustration only. An ordinarily skilled artisan will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

Overview

Disclosed is a system and related process for safety features in an active pen-stylus that protect the pen-stylus and extend its effective service life, according to an embodiment of the invention. These safety features pertain to both the front, marker tip end of the active pen-stylus and the rear, eraser end of the active pen-stylus. These safety features include a writing shaft impact shock absorber, a deformable, replaceable marker tip, a writing shaft safety stop, a writing shaft bottom bracket, and an eraser shaft impact shock absorber. A description of this specific invention begins at FIG. 15, following a description of a related pen-stylus that begins at FIG. 11. Before describing the invention in greater detail, a description will be provided of the e-paper tablet with which embodiments of the pen-stylus interact, as well as a description of an embodiment of the pen-stylus itself. Following this description, embodiments of the invention will be described.

Example System and Device Configuration—for Supporting Tablet Device

As shown in FIG. 1, an e-paper tablet device **110** receives inputs from the input mechanism **120**, for example, when the input mechanism **120** makes physical contact with a contact-sensitive surface (e.g., the touch-sensitive screen) on the e-paper tablet device **110** as the user makes a gesture of some sort with the input mechanism **120**. The input mechanism **120** may be a finger, pen-stylus or marker. The tablet device **110** here is referred to as an “e-paper tablet,” a device that mimics the feeling of writing with ordinary pen and paper for users of the device. Such devices are also known as “electronic paper” and “electronic ink.” Based on the nature of the contact, the e-paper tablet device **110** generates and executes instructions for updating content displayed on the contact-sensitive screen to reflect the gesture inputs. For example, in response to a gesture transcribing a verbal

5

message (e.g., a written text or a drawing), the e-paper tablet device 110 updates the contact-sensitive screen to display the transcribed message. As another example, in response to a gesture selecting a navigation option, the e-paper tablet device 110 updates the screen to display a new page associated with the navigation option. While embodiments of the invention have been designed for e-paper systems, embodiments of the invention may also be suitable for other forms of computing devices capable of receiving and processing inputs from pen-stylus devices.

The input mechanism 120 may refer to any device or object that is compatible with the contact-sensitive screen of the e-paper tablet device 110, in particular a pen-stylus device, such as a so-called active pen device having its own power source or a static pen that receives its power from engagement with the contact-sensitive screen on the e-paper tablet device 110. In one embodiment, the input mechanism 120 may work with an electronic ink (e.g., E-ink) contact-sensitive screen. For example, the input mechanism 120 may refer to any device or object that can interface with a screen and, from which, the screen can detect a touch or contact of said input mechanism 120. Once the touch or contact is detected, electronics associated with the screen generate a signal which the e-paper tablet device 110 can process as a gesture that may be provided for display on the screen. Upon detecting a gesture by the input mechanism 120, electronics within the contact-sensitive screen generate a signal that encodes instructions for displaying content or updating content previously displayed on the screen of the e-paper tablet device 110 based on the movement of the detected gesture across the screen. For example, when processed by the e-paper tablet device 110, the encoded signal may cause a representation of the detected gesture to be displayed on the screen of the e-paper tablet device 110, such as a scribble. As mentioned, the input mechanism 120 may be a pen-stylus or another type of pointing device, including a part of a user's body, such as a finger.

In one embodiment, the input mechanism 120 is an encased magnetic coil. When in proximity to the screen of the e-paper tablet device 110, the magnetic coil helps generate a magnetic field that encodes a signal that communicates instructions, which are processed by the e-paper tablet device 110 to provide a representation of the gesture for display on the screen, e.g., as a marking. The input mechanism 120 may be force (e.g., pressure) and tilt-sensitive such that the system can make natural, visual response to both the pressure and tilt applied by the user. In turn, the interaction between the input mechanism and the contact-sensitive screen of the e-paper tablet device 110 may generate a different encoded signal for processing, for example, to provide for display a representation of the gesture on the screen that has different characteristics, e.g., thicker line marking. In alternate embodiments, the input mechanism 120 includes a power source (e.g., a battery) which can generate an electric field with a contact-sensitive surface. It is noted that the encoded signal is a signal that is generated and may be communicated. The encoded signal may have a signal pattern that may be used for further analog or digital analysis (or interpretation).

In one embodiment, the contact-sensitive screen is a capacitive touchscreen. The screen may be designed using a glass or polymer material coated with a conductive material. Electrodes, or an alternate current carrying electric component, are arranged along the coating of the screen (e.g., in a diamond-shaped cross hatch) to maintain a constant level of current running throughout the screen. A second set of electrodes are arranged horizontally. The matrix of vertical

6

active electrodes and horizontal inactive electrodes generates an electrostatic field at each point on the screen. When an input mechanism 120 with conductive properties, for example the encased magnetic coil, a human finger, or something else that triggers the capacitive effect, is brought into contact with an area of the screen of the e-paper tablet device 110, current flows through the horizontally arranged electrodes, disrupting the electrostatic field at the contacted point on the screen. The disruption in the electrostatic field at each point that a gesture covers may be measured, for example as a change in capacitance, and encoded into an analog or digital signal.

In an alternate embodiment, the contact-sensitive screen is a resistive touchscreen. The resistive touch screen comprises two metallic layers: a first metallic layer in which striped electrodes are positioned on a substrate, such as a glass or plastic and a second metallic layer in which transparent electrodes are positioned. When contact from an input mechanism, for example a pen-stylus, finger, or palm, is made on the surface of the touchscreen, the two layers are pressed together. Upon contact, a voltage gradient is applied to the first layer and measured as a distance by the second layer to determine a horizontal coordinate of the contact on the screen. The voltage gradient is subsequently applied to the second layer to determine a vertical coordinate of the contact on the screen. The combination of the horizontal coordinate and the vertical coordinate register an exact location of the contact on the contact-sensitive screen. Unlike capacitive touchscreens which rely on conductive input mechanisms, a resistive touchscreen is configured to sense contact from nearly any input mechanism. Although some embodiments of the e-paper tablet device 110 are described herein with reference to a capacitive touchscreen, one skilled in the art would recognize that a resistive touchscreen could also be implemented.

In an alternate embodiment, the contact-sensitive screen is an inductive touchscreen. An inductive touchscreen comprises a metal front layer that is configured to detect deflections when contact is made on the screen by an input mechanism. Accordingly, an inductive touchscreen is configured to sense contact from nearly any input mechanism. Although some embodiments of the e-paper tablet device 110 are described herein with reference to a capacitive touchscreen, an ordinarily skilled artisan would recognize that alternative touchscreen technology may be implemented, for example, an inductive touchscreen could also be implemented.

The cloud server 130 is configured to receive information from the e-paper tablet device 110 and/or communicate instructions to the e-paper tablet device 110, according to some embodiments of the invention. As illustrated in FIG. 1, the cloud server 130 may comprise a cloud data processor 150 and a data store 160. Data recorded and stored by the e-paper tablet device 110 may be communicated via the network 140 to the cloud server 130 for storage in the data store 160. For example, the data store 160 may store documents, images, or other types of content generated or recorded by a user through the e-paper tablet device 110. In some embodiments, the cloud data processor 150 monitors the activity and usage of the e-paper tablet device 110 and communicates processing instructions to the e-paper tablet device 110. For example, the cloud data processor 150 may regulate synchronization protocols for data stored in the data store 160 with the e-paper tablet device 110.

Interactions between the e-paper tablet device 110 and the cloud server 130 are typically performed via the network 140, which enables communication between the e-paper

tablet device **110** and the cloud server **130**. In one embodiment, the network **140** uses standard communication technologies and/or protocols including, but not limited to, links using technologies such as Ethernet, 802.11, worldwide interoperability for microwave access (WiMAX), 3G, 4G, LTE, digital subscriber line (DSL), asynchronous transfer mode (ATM), InfiniBand, and PCI Express Advanced Switching. The network **140** may also utilize dedicated, custom, or private communication links. The network **140** may comprise any combination of local area and/or wide area networks, using both wired and wireless communication systems. The cloud server **130** may be alternatively implemented, and in some embodiments may be replaced by hardware and software that provide similar functionality while possibly not being considered a conventional cloud server.

FIG. 2 is a block diagram of the system architecture of an e-paper tablet device **110**, according to one example embodiment. In the embodiment illustrated in FIG. 2, the e-paper tablet device **110** comprises an input detector module **210**, an input digitizer **220**, a display system **230**, and a graphics generator **240**.

The input detector module **210** recognizes that a gesture has been or is being made on the screen of the e-paper tablet device **110**. The input detector module **210** refers to electronics integrated into the screen of the e-paper tablet device **110** that are configured to interpret an encoded signal generated by contact between the input mechanism **120** and the screen into a recognizable gesture. To do so, the input detector module **210** may evaluate properties of the encoded signal to determine whether the signal represents a gesture made intentionally by a user or a gesture made unintentionally by a user.

The input digitizer **220** may be configured to convert the analog signal encoded by the contact between the input mechanism **120** and the screen into a digital set of instructions. The converted digital set of instructions may be processed by the e-paper tablet device **110** to generate or update a user interface displayed on the screen to reflect an intentional gesture.

The display system **230** may include the physical and firmware (or software) components to provide for display (e.g., render) on a screen a user interface. The user interface may correspond to any type of visual representation that may be presented to or viewed by a user of the e-paper tablet device **110**.

Based on the digital signal generated by the input digitizer **220**, the graphics generator **240** may be configured to generate or update graphics of a user interface to be displayed on the screen of the e-paper tablet device **110**. The display system **230** may be configured to present those graphics of the user interface for display to a user using electronics integrated into the screen.

When an input mechanism **120** makes contact with a contact-sensitive screen of an e-paper tablet device **110**, the input detector module **210** recognizes a gesture has been made through the screen. The gesture may be recognized as a part of an encoded signal generated by a pressure or force sensor in the input mechanism **120** and/or corresponding electronics of the screen of the display system **230**. The encoded signal is transmitted to the input detector module **210**, which evaluates properties of the encoded signal in view of at least one gesture rule to determine whether the gesture was made intentionally by a user. If the input detector module **210** determines that the gesture was made intentionally, the input detector module **210** communicates the encoded signal to the digitizer output. The encoded

signal is an analog representation of the gesture received by a matrix of sensors embedded in the screen of the device **110**.

In one example embodiment, the input digitizer **220** translates the physical points on the screen that the input mechanism **120** made contact with into a set of instructions for updating what is provided for display on the screen. For example, if the input detector module **210** detects an intentional gesture that swipes from a first page to a second page, the input digitizer **220** receives the analog signal generated by the input mechanism **120** as it performs the swiping gesture. The input digitizer **220** generates a digital signal for the swiping gesture that provides instructions for the display system **230** of the e-paper tablet device **110** to update the user interface of the screen to transition from, for example, a current (or first page) to a next (or second page, which may be before or after the first page).

In one example embodiment, the graphics generator **240** receives the digital instructional signal, such as a swipe gesture indicating page transition (e.g., flipping or turning) generated by the input digitizer **220**. The graphics generator **240** generates graphics or an update to the previously displayed user interface graphics based on the received signal. The generated or updated graphics of the user interface are provided for display on the screen of the e-paper tablet device **110** by the display system **230**, e.g., displaying a transition from a current page to a next page to a user. In the displayed embodiment of the FIG. 2, the graphics generator **240** comprises a rasterizer module **250** and a depixelator module **260**. Input gestures drawn by a user on a contact-sensitive surface are received as vector graphics and are input to the rasterizer module **250**. The rasterizer module **250** converts the input vector graphics to raster graphics, which can be displayed (or provided for display) on the contact-sensitive surface. The depixelator module **260** may apply image processing techniques to convert the displayed raster graphics back into vector graphics, for example to improve processing power of the e-paper tablet device **110** and to conserve memory of the e-paper tablet device **110**. In at least one implementation, the depixelator module **260** may convert a displayed raster graphic back to a vector graphic when exporting content displayed on the screen into a different format or to a different system.

Further details about structures and functions of e-paper tablets and their graphical displays can be found in U.S. Pat. No. 11,158,097 to Martin Sandsmark and Gunnar Sletta entitled "Generating vector graphics by processing raster graphics" and in U.S. Pat. No. 10,824,274 to Sondre Hoff Dyvik, Martin Sandsmark, and Magnus Haug Wanberg, entitled "Interactive displays," both of which are incorporated by reference herein.

FIG. 3 illustrates a front and right perspective view of an e-paper tablet **300** having the functionality described for the e-paper tablet device **110** in FIGS. 1-2. Among other things, the e-paper tablet **300** includes a touch-sensitive display **303**. The display **303** has been treated to provide a paper-feeling for users of the device when they engage with it using an input device **120**. FIG. 3 also shows a charging area **304** for recharging the input device **120**, when the input device is an active pen-stylus, according to an embodiment of the invention. Inside the e-paper tablet **300** near where the charging area **304** is located may be a set of magnets to hold the input device **120** in place while it is re-charging. FIG. 3 also shows a USB-c connector **307** that may be used to provide electrical power to the e-paper tablet **300**, as well as transmitting various types of data into or out of the e-paper tablet **300**.

The e-paper tablet **300** also includes several actuators and other features that will be shown below in FIGS. **5-10**.

FIG. **4** illustrates hardware components of an example Electrophoretic Display (EPD) in accordance with a disclosed embodiment. As discussed, a variety of display technologies may be employed, including EPDs, LCDs, and reflective LCDs (rLCDs). The specific display device deployed may be part of the display system **230** of the e-paper tablet device **110** shown in FIG. **2** and produce the images shown on the display **303** of the e-paper tablet **300** shown in FIG. **3**. The EPD includes a gate driver **409**, a source driver **411**, a shift register **423** with data and clock signal line, a latch **425**, a voltage selector **427**, and rows making up a display **405**. The EPD industry borrowed certain components and concepts from the LCD industry; however, these two devices have some fundamental differences as well. Of particular relevance here is the persistence of pixels in EPD displays. Unlike LCD displays, EPD displays do not require the frequent refreshing required in an LCD display. In an EPD display, once a neutral voltage is set for a pixel, the pixel will not change, for example, and will persist for a long period of time, especially relative to an LCD display.

As mentioned, Electrophoretic displays (EPDs) **405** have utilized many aspects of LCD production infrastructure and driving mechanisms. The driving electronics typically consist of a gate driver (GD) **409** and a source driver (SD) **411**. The display **405** has multiple rows of pixels. Pixel values within a row may be changed, e.g., logic high voltage may be a “black” pixel and a logic low voltage or “ground” may be a no color pixel. The pixels in the EPD **405** function similarly to small capacitors that persist over long time intervals. An EPD pixel contains a large number of charged particles that are suspended in a liquid. If a charge is applied, the particles will move to a surface where they become visible. White and black particles have opposite charges such that a pixel’s display may change from white to black by applying an opposite charge to the pixel. Thus, the waveforms applied to an EPD comprise long trains of voltages to change from black to white or vice versa. The EPD arts are also known to have the ability to apply variable voltage levels that mix the white and black particles to produce various shades of gray. Voltage levels in a pixel also may be tiered between to provide shades between no color and black (e.g., levels of grey). Groups of pixels around each other may form a region that provides some visible characteristic to a user, e.g., an image on a screen, e.g., of the display system **230** of the e-paper tablet device **110**.

To change pixel values in a region, a scan of a display **405** will conventionally start at a top row, e.g., row **0** **421**, and apply voltages to update pixels within a particular row where pixels need to be changed to correspond with the image that is displayed. In this example, a start pulse (GDSP) **403** can be used to reset the driver **411** to row **0** **421**, and a direction (DIR) **404** can be used to reset a direction. A row-by-row selection is made by driving the driver gate **409** to select a row, e.g., active row **413**. All pixels in one row are addressed concurrently using data transferred to the display. Latch **425** receives from the shift register **423** the next set of voltages to be applied to a row of pixels. When the scan of the active row is completed and, if necessary, pixels changed or updated, a clock pulse (GDCLK) **415** is issued to the driver gate **409** to change to the next row **417** for a scan.

As mentioned above, an ordinary artisan will recognize that a similar function can be accomplished also with a standard LCD, OLED, MicroLED or other type of display,

and the description of EPD technology is provided here merely for illustration of one embodiment of the invention.

The source driver **411** is used to set the target voltage for each of the pixels/columns for the selected row. It consists of a shift register **423** for holding the voltage data, a latch circuit **425** for enabling pixel data transfer while the previous row is being exposed, and a voltage selector (multiplexer) **427** for converting the latched voltage selection into an actual voltage. For all rows to be updated all the voltage values have to be shifted into the register **423** and latched for the voltages to be available.

FIG. **5** is a block diagram illustrating components of an example machine able to read instructions from a machine-readable medium and execute them in a processor (or controller), according to one embodiment. In this example, FIG. **5** shows a diagrammatic representation of a machine in the example form of a computer system **500** (e.g., the computing portions of the e-paper tablet **111** shown in FIG. **1**) within which program code (e.g., software) for causing the machine to perform any one or more of the methodologies discussed herein may be executed. The e-paper tablet device **110** may include some or all of the components of the computer system **500**. The program code may be comprised of instructions **524** executable by one or more processors **502**. In the e-paper tablet system **110**, the instructions may correspond to the functional components described in FIGS. **1**, **2**, and **4**. FIG. **5** is an example of a processing system, of which a some of the described components or all of the described components may be leveraged by the modules described herein for execution.

While the embodiments described herein are in the context of the e-paper tablet system **110**, it is noted that the principles may apply to other touch sensitive devices. In those contexts, the machine of FIG. **5** may be a server computer, a client computer, a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a cellular telephone, a smartphone, a web appliance, a network router, an internet of things (IoT) device, a switch or bridge, or any machine capable of executing instructions **524** (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute instructions **524** to perform any one or more of the methodologies discussed herein.

The example computer system **500** includes one or more processors **502** (e.g., a central processing unit (CPU), one or more graphics processing units (GPU), one or more digital signal processors (DSP), one or more application specific integrated circuits (ASICs), one or more radio-frequency integrated circuits (RFICs), or any combination of these), a main memory **504**, and a static memory **506**, which are configured to communicate with each other via a bus **508**. The computer system **500** may further include visual display interface **510**. The visual interface may include a software driver that enables displaying user interfaces on a screen (or display). The visual interface may display user interfaces directly (e.g., on the screen) or indirectly on a surface, window, or the like (e.g., via a visual projection unit). For ease of discussion the visual interface may be described as a screen or display screen. The visual interface **510** may include or may interface with a touch enabled screen, e.g., of the e-paper tablet system **110** and may be associated with the display system **230**. The computer system **500** may also include an input device **512** (e.g., a pen-stylus, a keyboard, or touch screen keyboard), a cursor control device **514** (e.g., a mouse, a trackball, a joystick, a motion sensor, or other

pointing instrument), a storage unit **516**, a signal generation device **518** (e.g., a speaker), and a network interface device **520**, which also are configured to communicate via the bus **508**.

The storage unit **516** includes a machine-readable medium **522** on which is stored (or encoded) instructions **524** (e.g., software) embodying any one or more of the methodologies or functions described herein. The instructions **524** (e.g., software) may also reside, completely or at least partially, within the main memory **504** or within the processor **502** (e.g., within a processor's cache memory) during execution thereof by the computer system **500**, the main memory **504** and the processor **502** also constituting machine-readable media. The instructions **524** (e.g., software) may be transmitted or received over a network **426** via the network interface device **520**.

While machine-readable medium **522** is shown in an example embodiment to be a single medium, the term "machine-readable medium" should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions (e.g., instructions **524**). The term "machine-readable medium" shall also be taken to include any medium that is capable of storing instructions (e.g., instructions **524**) for execution by the machine and that cause the machine to perform any one or more of the methodologies disclosed herein. The term "machine-readable medium" includes, but not be limited to, data repositories in the form of solid-state memories, optical media, and magnetic media.

The computer system **500** also may include the one or more sensors **525**. Also note that a computing device may include only a subset of the components illustrated and described with FIG. **5**. For example, an IoT device may only include a processor **502**, a small storage unit **516**, a main memory **504**, a visual interface **510**, a network interface device **520**, and a sensor **525**.

Representative e-Paper Tablet

FIG. **3** provided a representative view of an e-paper tablet **300**, resembling the e-paper **110** shown in FIGS. **1-2**. FIG. **6** illustrates a rear view of the e-paper tablet **300** showing volcano feet **601a-601d**, a pogo pad **603**, and an antenna region **605**, according to an embodiment of the invention. The antenna region **605** resides outside and above the location for a main antenna (e.g., an antenna on the e-paper tablet **110** that communicates with the cloud server **130** shown in FIG. **1** and which may generate the e-paper tablet's beacon signal discussed below with respect to FIG. **14**) for the e-paper tablet **300**, allowing the e-paper tablet device **300** to connect to the Internet, for example. The pogo pad **603** allows the e-paper tablet device **300** to connect to other devices, such as a folio device having a keyboard, for example.

FIG. **7** illustrates a top view of the e-paper tablet device **300** showing volcano feet **601a**, **601d**, and a power button **701**, according to an embodiment of the invention.

FIG. **8** illustrates a bottom view of the e-paper tablet device **300** showing volcano feet **601b**, **601d** and the USB-c connector **307**, according to an embodiment of the invention.

FIG. **9** illustrates a right view of the e-paper tablet device **300** showing volcano feet **601b**, **601a**, and the charging area **304** for recharging the input device **120**, when the input device is an active pen-stylus, according to an embodiment of the invention. Inside the e-paper tablet **300** near where the charging area **304** is located may be a set of magnets to hold the input device **120** in place while it is re-charging.

FIG. **10** illustrates a left view of the e-paper tablet device **300** showing volcano feet **601c**, **601d**, according to an embodiment of the invention.

Example Active Pen in Operation

An active pen-stylus (or more commonly "active pen") is a pen-stylus input device that allows users to e.g., write, sketch or draw on the display of the computing device, e.g., the e-paper tablet **110**. An active pen-stylus includes digital components and/or circuitry that communicate with the computing device, e.g., the e-paper tablet. This communication enables advanced features such as force (e.g., pressure) sensitivity, tilt detection, programmable buttons, palm detection, eraser tips, memorizing settings, and writing data transmission. Viewed more expansively, communications between the computing device and the active pen-stylus enables a wide mix of peripheral sensors to be placed in the active pen-stylus with the resulting data reported to the computing device, e.g., the e-paper tablet. Such sensors placed in the active pen-stylus may range from simple buttons to enhanced artificial intelligence features.

An active pen's electronic components typically include a power source that may enable the device's electronics to provide lower latency and greater fidelity than other pen types, e.g., a passive pen. Active pens provide a number of advantages over passive pens, including hover latency, e.g., an active pen may typically be activated by merely being in proximity to a display, e.g., the display associated with the e-paper tablet **110**.

Once the active pen touches or contacts a display screen of a device like an e-paper tablet, electronics associated with the display screen generates a signal which the e-paper tablet (e.g., the e-paper tablet **110**) can process as a gesture made by the user. Upon detecting the gesture by the pen-stylus, electronics within the contact-sensitive screen generates a signal that encodes instructions for displaying content or updating content previously displayed on the screen of the e-paper tablet device based on the movement of the detected gesture across the screen.

In contrast with an active pen, a passive pen typically has no internal power source. A passive pen remains in an inactive state until the pen touches a device screen (e.g., a tablet device screen) causing a signal to pass from the device through the passive stylus and back to the device. The electronics associated with a passive pen may be integrated into the pen-stylus device or even provided in a small cartridge that is placed inside a pen-shaped stylus cover designed to better suit human ergonomics than the small cartridge containing the electronics and other components.

FIG. **11** illustrates an active pen-stylus **1100** that comprises a core member **1102** that itself comprises one or more antennas configured for communications with a tablet device, such as the e-paper tablet **110**. The active pen-stylus **1100** may include one or more force sensing systems **1104** that detect force, e.g., the forces exerted on a display of an e-paper tablet **110** by a user. In addition, the active pen-stylus **1100** may also help emulate human interaction with conventional writing utensils like pencil and paper, e.g., render "a pencil and paper stack feeling". The active pen-stylus **1100** also includes a power source, e.g., a battery **1106**. Among other things, the battery **1106** allows the active pen-stylus **1100** to support a "hover" function that allows the active pen-stylus **1100** to enter a sleep state for battery charge preservation when it is not actively engaged with the display of the e-paper tablet **110** and to wake up from the sleep state when the core member **1102** detects a proximity to the display of the e-paper tablet **110**. The active pen-stylus **1100** might not actually draw lines on the display of the

13

e-paper tablet **110** until a tip of the active pen-stylus **1100** physically touches the display of the e-paper tablet **110**, according to an embodiment of the invention.

An active pen **1100** conventionally comprises a PCBA **1105** which includes electronic components needed for driving the signal lines associated with the core member **1102**. The PCBA **1105** could alternatively be provided as a flexible printed circuit (FPC). The PCBA **1105** may include an appropriate active pen PCBA or IC/ASIC/MCU that processes data received from the core member **1104** for sensing force or displacement pressure with high resolution sensing of the force during the interaction between the active pen **1100** and an object, e.g., the surface of the computing device, such as the e-paper tablet **110**.

The active pen **1100** may serve as the input mechanism **120** described in FIG. 1 that provides data input to the e-paper tablet **110**, which may result in a drawing appearing on the display of the e-paper tablet **110**. The active pen **1100** may also include an erasure system that receives user instructions related to erasing portions of a display on the screen of the associated computing device, e.g., the e-paper tablet **110**. In a sense, the erasure system operates identically as the active pen-stylus system but where one draws, the other erases.

In operation, the force sensing system **1104** receives physical forces imparted to a marker tip of the active pen **1100** (e.g., the marker tip **1202** shown in FIG. 12) when the marker tip engages with a display on a tablet device (e.g., the e-paper tablet **110**) and translates the physical force received into an electronic signal that is transmitted to the PCBA **1105** whose electronic components may perform a variety of processes on the signals received. The PCBA **1105** may then transmit the signals back to the tablet device (e.g., the e-paper tablet **110**) for further action (e.g., drawing a line).

In embodiments where a secondary antenna system in the active pen **1100** enables tail eraser functionality, the active pen **1100** also comprises a second antenna system **1103**, which enables the active pen **1100** to transmit and receive signals with the computing device (e.g., the e-paper tablet **110**) through the tail eraser portion of the active pen **1100** near a cap **1108**. In this embodiment, the active pen **1100** also includes a second force (e.g., pressure) sensing system **1107** that controls the force (e.g., pressure) imparted to the display of the computing device from the erasure activity. Likewise, the PCBA **1105** may include an appropriate active pen PCBA or IC/ASIC/MCU that process data associated with erasure functionality received from the second antenna system **1103**.

The active pen **1100** also typically includes an external casing **1101** as a pen-stylus holder, typically formed in a cylindrical shape and made of non-metal material such as a plastic that contains the internal electronics within the casing **1101**. The top end of the casing **1101** may be provided with the cap **1108**.

FIG. 12 illustrates an external casing **1201** for a pen-stylus **1200** designed to fit a user's hand, according to an embodiment of the invention. The pen-stylus **1200** includes a marker tip **1202** and an eraser **1203**. The marker tip **1202** operates as a part of the core member **1102** shown in FIG. 11, and the eraser **1203** operates as part of the second antenna system **1103** shown in FIG. 11.

When an active pen-stylus (e.g., the pen-stylus **1100** and/or the pen-stylus **1200**) includes multiple antennas as part of the core (e.g., the core **1102**), then the pen-stylus may provide additional capabilities. As shown in FIGS. 13A-13B, employing two separate transmitters **1301**, **1303** in the marker tip (e.g., the core **1102** shown in FIG. 11) of the

14

active pen-stylus **1308** gives the computing device (e.g., the e-paper tablet **110**) the ability to measure two distinct signals (illustrated in FIG. 13A as **1301a**, **1303a** and illustrated in FIG. 13B as **1301b**, **1303b**). By knowing the antenna separation in the input device (e.g., the active pen-stylus **1308**), the input digitizer **220** in a larger computing device, such as the e-paper tablet **110**, can derive the active pen-stylus tilt angle θ **1307** relative to the computing device (e.g., the e-paper tablet **110**). The input digitizer **220** is conventionally able to process active pen-stylus tilt information related to the drawing portion of the active pen-stylus. Only a slight change needs to be made to the input digitizer **220** for it to process tilt angle data related to an eraser (if erasure functionality is provided) to cause a change to the display on the e-paper tablet **110**.

As seen in FIG. 13A, the signals **1301a**, **1303a** from the two antennas **1301**, **1303** coincide when the active pen-stylus **1308** is perpendicular to the graphics display of the computing device (as shown by the grid **1305** and the line passing through the centers of both signals **1301a**, **1303a**). In contrast, FIG. 13B illustrates the active pen-stylus **1308** tilted by an angle θ **1307** relative to the display of the computing device (e.g., the e-paper tablet **110**) as shown by the grid **1305** and the distance D marking the distance between the two centers of signals **1301b**, **1303b**. With such a tilt, the signals **1301b**, **1303b** from the two antennas **1301**, **1303** are not aligned and are separated by the distance D . This distance D may be used to determine the corresponding drawing amount performed by the e-paper tablet device **110** on the display.

The method for deriving the tilt angle θ **1307** of the active pen-stylus **1308** is known when using two antennas (e.g., two separate antennas in the antenna system **1102** shown in FIG. 11, presented in FIGS. 13A and 13B as antennas **1301**, **1303**). The distance between antenna **1301** and antenna **1303** is fixed and known. This known separation, combined with basic trigonometry, can be employed in active pen tip antenna systems to derive the tilt angle for the active pen-stylus tip (e.g., the angle of the tip of the core member **1102** shown in FIG. 11 relative to the display **230** shown in FIG. 2 on the e-paper tablet **110**). Thus, in one embodiment of the invention, as the tilt angle θ increases, the shading area proportionately increases on the display. One could similarly imagine a slightly different arrangement of the components such that as the tilt angle θ decreased, the area of marking area proportionately increased, e.g., in inverse proportion. In both embodiments, the area of marking still has a direct correlation to the tilt angle θ , e.g., as the tilt angle changes, the area of marking changes.

This tilt angle determination would operate in a similar manner for an erasure function. Embodiments of the invention allow for the derivation of the tilt angle of the tail eraser by employing an electronically calculated method in the input digitizer **220** similar to that employed conventionally by the input digitizer **220** for determining the tilt of the marker tip and the eraser function, when present. Such a calculation requires that the pen-stylus provide additional data/information to the input digitizer **220**.

In some embodiments, the pen-stylus integrated circuits (e.g., included in the PCBA **1105**) may not have a sufficient number of antenna signal lines for driving two transmitters in both the active pen tip and tail eraser antenna system. Conventional active pens have two antenna signal lines for the active pen-stylus tip and one antenna signal line for a tail eraser. Thus, no more than three antenna signal lines. With this conventional configuration, the input digitizer **220** (or comparable hardware) cannot derive the tilt information for

15

the tail eraser. This problem has been solved in a manner that allows a conventional active pen configuration to drive four antennas rather than the conventional three antennas, thus enabling enhanced erasure capabilities. Further details about structures and functions for solving this problem may be found in U.S. application Ser. No. 18/208,280 to Gaute Nordby et al., entitled "Active Pen-Stylus Precise Eraser," which is incorporated by reference herein.

FIG. 14 illustrates a cross section of the forward portion of an active pen-stylus 1400 that a user may employ for tasks such as drawing a line on the display of a device (e.g., the display 303 of the e-paper tablet 300 shown in FIG. 3), according to an embodiment of the invention. The active pen-stylus 1400 includes a core (e.g., the core member 1102 shown in FIG. 11) comprised of components such as a first antenna 1407 and a second antenna 1425. The active pen-stylus 1400 also includes a force sensor (e.g., the force sensing system 1104 shown in FIG. 11) comprised of components such as a writing shaft 1413, a first spring coil 1431, a second spring coil 1429, and a force sensor 1423.

The active pen-stylus 1400 provides high accuracy for various drawing tasks due to interactions between the first antenna 1407 and the second antenna 1425, in part because of the placement of the first antenna 1407 and the second antenna 1425 with respect to each other and with respect to the active pen-stylus 1400 itself. As shown in FIG. 14, the second antenna 1425 surrounds the first antenna 1407, but the second antenna 1425 and the first antenna 1407 do not overlap horizontally in the active pen-stylus 1400 and are further separated from each other by the insulator 1405.

Each antenna 1407, 1425 may send a clear signal to the e-paper tablet 110. This arrangement is particularly helpful for the antenna 1425 since the antenna 1407 is often in physical contact with the display of the e-paper tablet 110. When the pen-stylus 1400 is not in contact (e.g., out of range) with the tablet (e.g., the e-paper tablet 110), antennas on the tablet periodically send a beacon signal. This beacon signal is meant for the pen-stylus 1400 to detect that it is within range of the tablet. In a time following the beacon signal, the e-paper tablet 110 typically keeps an open time slot for the pen-stylus 1400 to acknowledge the beacon and respond to the e-paper tablet 110. This acknowledgement initiates the two-way communication between the pen-stylus 1400 and the tablet (e.g., the e-paper tablet 110). As an ordinary artisan will recognize, the "beacon" from the tablet is a form of an uplink to the pen-stylus 1400, and the communication from the pen-stylus 1400 to the tablet is a form of a downlink.

The active pen-stylus 1400 also offers advanced force (e.g., pressure) sensing, e.g., a low activation force, high maximum force, and a high dynamic range. To translate the detecting force imparted by the pen-stylus 1400 against the display of the e-paper tablet 110, the writing shaft 1413 is designed to move towards and/or away from a force sensor 1423, which registers activity between a replaceable marker tip 1403 and the display (e.g., the display of the e-paper tablet 110), including the pressure applied by the user of the pen-stylus 1400 to the display. Among other things, a bottom bracket 1419 prevents the writing shaft 1413 from rotating about its center axis. As previously discussed, the writing shaft 1413 is designed to move towards and/or away from a force sensor 1423, which registers activity between a replaceable marker tip 1403 and the display (e.g., the display of the e-paper tablet 110), including the force (e.g., pressure) applied by the user of the pen-stylus 1400 to the display. The active pen-stylus 1400 has been designed to have a minimum set of components, according to an embodiment of the

16

invention. Further details about structures and functions of active pen-stylus writing systems can be found in U.S. application Ser. No. 18/779,164, filed on Jul. 22, 2024 entitled "Marker Writing System" and given reference number "21149403," which is incorporated by reference herein.

The active pen-stylus 1400 includes a replaceable marker tip 1403 that includes the core antenna 1407 and an insulator 1405, according to an embodiment of the invention. The replaceable marker tip 1403 is held to the writing shaft 1413 by crush ribs 1409 and designed for hand removal by the user. The replaceable marker tip 1403 has a variable lifetime, depending on factors such as user personal sensitivity and user use patterns. Further details about structures and functions of replaceable marker tips can be found in U.S. application Ser. No. 18/779,151, filed on Jul. 22, 2024 entitled "Replaceable Conductive Marker Tip," and given reference number "21149400," which is incorporated by reference herein.

As discussed further below, the active pen-stylus 1400 offers a variety of device safety features for its operational protection and long service life. Among other things, the writing shaft 1413 may be fitted with an impact shock absorber 1417 that is designed to receive heavy forces imparted to the pen-stylus 1400 (e.g., the force arising from being dropped onto a hard surface) so that an upper limit is set for the forces transmitted to the force sensor 1423.

The replaceable marker tip 1403 has also been designed to leave a marker tip clearance 1411 between itself and the marker body 1401, according to an embodiment of the invention. The marker tip clearance 1411 may also be helpful in receiving certain shock forces to the pen-stylus 1400 that might otherwise be received by components inside the pen-stylus 1400. In the event of a severe shock, the replaceable marker tip 1403 may also function as a sacrificial element that may even absorb otherwise destructive energy. In situations where the replaceable marker tip 1403 has absorbed so much destructive energy that it can no longer function, then the user of the pen-stylus 1400 may simply need to add a new replaceable marker tip 1403 to an otherwise workable pen-stylus 1400.

The pen-stylus 1400 may include other components, such as a writing system spring 1421 which in cooperation with the force sensor 1423 may present to the user of the pen-stylus 1400 the feeling of writing on a stack of paper. Further details about structures and functions of active pen-stylus paper emulation can be found in U.S. application Ser. No. 18/779,154, filed on Jul. 22, 2024 entitled "Advanced Paper Emulation," and given reference number "21149401," which is incorporated by reference herein.

Pen-Stylus Device Safety Elements

As mentioned, the active pen-stylus 1400 includes a variety of safety features for its operational protection and to extend its effective service life, according to an embodiment of the invention. These safety features pertain to both the front, marker tip end of the active pen-stylus 1400 and the rear, eraser end of the active pen-stylus 1400. These safety features include:

- A writing shaft impact shock absorber 1417,
- A deformable, replaceable marker tip 1403,
- A writing shaft safety stop 1427,
- A writing shaft bottom bracket 1419, and
- An eraser shaft impact shock absorber 1611 (shown in FIG. 16)

FIG. 15 provides an exploded view of the forward end of the pen-stylus 1400 shown in FIG. 14 and particularly points out various device safety features found in the forward section of the pen-stylus 1400, namely, the writing shaft

17

impact shock absorber **1417**, the deformable, replaceable marker tip **1403**, the writing shaft safety stop **1427**, and the writing shaft bottom bracket **1419**, according to an embodiment of the invention.

The writing shaft **1413** may be fitted with an impact shock absorber **1417** that is designed to receive heavy or excess forces imparted to the pen-stylus **1400** (e.g., the force arising from the pen-stylus **1400** being dropped onto a hard surface) so that fewer of these excess forces are transmitted to the force sensor **1423**, which is often particularly susceptible to damage. Without the impact shock absorber **1417**, if the pen-stylus **1400** is dropped accidentally and with sufficient excess force, then it is possible for the force sensor **1423** to short, disrupting its ability to function properly in the task of translating mechanical input forces received from the writing shaft **1413** into electrical ones for operation by the electronics of the pen-stylus **1400** (e.g., the PCBA **1105** and its electronic components shown in FIG. **11**). Accordingly, the impact shock absorber **1417** has been designed to absorb shocks (excess forces) imparted to the pen-stylus **1400**, particularly shocks (excess forces) received along the writing shaft **1413**, effectively increasing the threshold above which the force sensor **1423** may malfunction due to a received shock.

As another safety feature, the replaceable marker tip **1403** has also been designed to be a deformable part that absorbs impact energy (excess forces) and may even be a sacrificial element, according to an embodiment of the invention. Thus, if a user drops the pen-stylus **1400** in an orientation that causes it to hit an object on the front tip end of the pen-stylus **1400**, then the replaceable marker tip **1403** may absorb some of the energy (excess forces), even including accepting deformation to its shape.

In particularly severe blows, the replaceable marker tip **1403** might even be sheared off at the safety gap **1411**, according to an embodiment of the invention. Thus, the replaceable marker tip **1403** may function as a sacrificial element for absorbing excess forces. In such cases, the user of the pen-stylus **1400** would simply need to add a new replaceable marker tip **1403** to an otherwise workable pen-sensor **1400**. Thus, the user could simply replace the broken replaceable marker tip **1403** with a new replaceable marker tip.

The replaceable marker tip **1403** is held on by force to the writing shaft **1413**. The replaceable marker tip **1403** has also been designed not to rotate on the writing shaft **1413**. The crush ribs **1409** hold the core antenna **1407** onto the writing shaft **1413**. As mentioned, the replaceable marker tip **1403** is designed to be removable. In particular, some embodiments of the invention allow the replaceable marker tip **1403** to be hand removable by users without the need for tools. When users remove the replaceable marker tip **1403** to insert a new marker tip **1403**, the writing shaft **1413** will receive an increase in force (e.g., pressure) that arises from the user effectively tugging on the end of the writing shaft **1413**. If a user exerts sufficient force (e.g., excess force) on the writing shaft **1413**, it would be possible for the user to inadvertently extract the writing shaft **1413** from the pen-stylus **1400**, which might cause the pen-stylus **1400** to malfunction. The positive stop **1427** functions as a writing shaft safety stop, preventing the writing shaft **1413** from being removed from the pen-stylus **1400**. The positive stop **1427** may be included in a writing shaft captivator **1415**. Thus, the positive stop **1427** acts as a safety feature to protect the pen-stylus **1400**.

As a final safety feature for the forward end of the pen-stylus **1400**, a bottom bracket **1419** holds the writing

18

shaft **1413** to prevent the writing shaft **1413** from rotating along its axis, which might occur as the user imparts forces (e.g., excess force) to the replaceable marker tip **1403**, especially given how the replaceable marker tip **1403** attached to the writing shaft **1413**.

The writing system of the pen-stylus **1400** may also include erasure functionality, according to an embodiment of the invention. An Inertial Measurement Unit (IMU) (not shown) in FIG. **14** in the pen-stylus **1400** senses the orientation of the pen-stylus **1400** and engages the electronics in the pen-stylus **1400** (e.g., the PCBA **1105** shown in FIG. **11**) to flip to the eraser function as the pen rotates away from the writing function. A precise eraser invention has already been created by the inventors of the present invention and may be found in U.S. application Ser. No. 18/208,280 to Gaute Nordby et al., entitled "Active Pen-Stylus Precise Eraser," which is incorporated by reference herein.

Just as the forward end of the pen-stylus **1400** includes various safety features, the rear end of the pen-stylus **1400** includes various safety elements to protect the components associated with the erasure functionality from excess forces, according to an embodiment of the invention.

FIG. **16** illustrates a rear portion **1600** of the pen-stylus **1400** that includes an eraser cap **1601**, a first rear antenna **1603**, a second rear antenna **1607**, a rear erasure shaft **1605**, a rear eraser gap **1615**, a rear impact shock absorber **1611**, and a rear force sensor **1609**, according to an embodiment of the invention.

The rear impact shock absorber **1611** provides similar safety functions as its counterpart on the front end of the pen-stylus **1400**. In other words, the rear impact shock absorber **1611** provides a similar function for the erasure system that the impact shock absorber **1417** provides for the writing system. The rear force sensor **1609** provides the same function for the erasure system that the force sensor **1423** provides for the writing system.

An end erasure cap **1601** may provide a similar deformable structure to the replaceable marker tip **1403**, according to an embodiment of the invention. If the body of the pen-stylus **1400** receives excess forces along certain angles at a rear of the body, the end erasure safety cap absorbs an amount of the excess force, shielding the pen-stylus **1400** from the shearing force.

The first rear antenna **1603** and the second rear antenna **1607** provide similar functionality in support of the erasure system as the front antenna **1407** and the front antenna **1425** provide for the writing system. The rear eraser shaft **1605** provides similar functionality for the erasure system that the writing shaft **1413** provides for the writing system.

Additional Considerations

As discussed below, these pen-styluses can be equipped with a tail eraser allowing the user to erase content from the display of the computing device, e.g., the e-paper tablet **110**. An active pen includes electronics components which enables the active pen to send and receive signals from the computing device.

This disclosed configuration provides additional precision and options for users as they go about erasing portions of drawings on an e-paper tablet. This should improve the efficiency of users interacting with e-paper tablets while also enabling them with more precise functional capabilities.

It is to be understood that the figures and descriptions of the present disclosure have been simplified to illustrate elements that are relevant for a clear understanding of the present disclosure, while eliminating, for the purpose of clarity, many other elements found in a typical system. Those of ordinary skill in the art may recognize that other

19

elements and/or steps are desirable and/or required in implementing the present disclosure. However, because such elements and steps are well known in the art, and because they do not facilitate a better understanding of the present disclosure, a discussion of such elements and steps is not provided herein. The disclosure herein is directed to all such variations and modifications to such elements and methods known to those skilled in the art.

Some portions of above description describe the embodiments in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as engines, without loss of generality. The described operations and their associated engines may be embodied in software, firmware, hardware, or any combinations thereof.

As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the invention. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise. While particular embodiments and applications have been illustrated and described, it is to be understood that the disclosed embodiments are not limited to the precise construction and components disclosed herein. Various modifications, changes and variations, which will be apparent to those skilled in the art, may be made in the arrangement, operation and details of the method and apparatus disclosed herein without departing from the spirit and scope defined in the appended claims.

The invention claimed is:

1. A pen-stylus, comprising:

a marker tip that engages with a display of a tablet device to produce mechanical input forces as a user engages the pen-stylus on the display of the tablet device and transmits the mechanical input forces;
a writing shaft that receives the mechanical input forces transmitted from the marker tip;

20

a force sensor that translates the mechanical input forces received from the writing shaft into electrical forces for operation by electronics in the pen-stylus;

a writing shaft impact shock absorber that receives excess forces imparted to the pen-stylus to shield the force sensor from receiving the excess forces; and

wherein the marker tip comprises deformable materials and wherein when a body of the pen-stylus receives a shearing force along angles at a front of the body of the pen-stylus that include the marker tip, the marker tip absorbs at least some of the shearing force by deforming its shape.

2. The pen-stylus of claim 1, wherein the excess forces arise from the pen-stylus being dropped onto a hard surface.

3. The pen-stylus of claim 1 wherein the marker tip shears off when the shearing force exceeds a predetermined threshold, shielding the pen-stylus from the shearing force.

4. The pen-stylus of claim 3 wherein the body of the pen-stylus includes a front marker safety gap between the marker tip and the body of the pen-stylus, wherein the front marker safety gap provides a shearing off location for the marker tip.

5. The pen-stylus of claim 1 wherein the marker tip is replaceable.

6. The pen-stylus of claim 1, further comprising:

a writing shaft bottom bracket that holds the writing shaft in place to prevent the writing shaft from rotating along its axis.

7. The pen-stylus of claim 6,

wherein the marker tip is removeable and wherein the writing shaft bottom bracket prevents dislocation of the writing shaft when the user imparts excessive force to the writing shaft while changing the removeable marker tip.

8. The pen-stylus of claim 1, wherein the tablet device is an e-paper tablet device wherein interaction between the display and the pen-stylus emulates conventional paper in sensations felt by the user of the pen-stylus.

9. The pen-stylus of claim 8, wherein the e-paper tablet device sends a beacon signal to the pen-stylus to establish a communications uplink with the pen-stylus, which causes the pen-stylus to return a communications downlink.

10. A pen-stylus, comprising:

a removable marker tip that engages with a display on a tablet device as a user engages the pen-stylus on the display of the tablet device and transmits mechanical input forces, wherein the removable marker tip is designed for hand removal;

a writing shaft that receives the mechanical input forces from the removable marker tip as the user engages the pen-stylus on the display of the tablet device;

a force sensor that translates the mechanical input forces received from the writing shaft into electrical forces for operation by electronics in the pen-stylus;

a writing shaft impact shock absorber that receives excess forces imparted to the pen-stylus to shield the force sensor from receiving the excess forces; and

a writing shaft safety stop that prevents dislocation of the writing shaft when the user imparts excessive force in changing the removeable marker tip.

11. The pen-stylus of claim 10, wherein the excess forces arise from the pen-stylus being dropped onto a hard surface.

12. The pen-stylus of claim 10, further comprising:

a writing shaft bottom bracket that holds the writing shaft in place to prevent the writing shaft from rotating along its axis.

21

13. The pen-stylus of claim 12, wherein the writing shaft bottom bracket prevents dislocation of the writing shaft when the user imparts excessive force to the writing shaft while changing the removeable marker tip.

14. The pen-stylus of claim 10, wherein the tablet device is an e-paper tablet device wherein interaction between the display and the pen-stylus emulates conventional paper in sensations felt by the user of the pen-stylus.

15. The pen-stylus of claim 14, wherein the e-paper tablet device sends a beacon signal to the pen-stylus to establish a communications uplink with the pen-stylus, which causes the pen-stylus to return a communications downlink.

16. The pen-stylus of claim 10 wherein the marker tip shears off when a shearing force exceeds a predetermined threshold, shielding the pen-stylus from the shearing force.

17. The pen-stylus of claim 16 wherein the body of the pen-stylus includes a front marker safety gap between the marker tip and the body of the pen-stylus, wherein the front marker safety gap provides a shearing off location for the marker tip.

18. A pen-stylus, comprising:

- a writing shaft that receives mechanical input forces as a user engages the pen-stylus on a display of a tablet device;
- a force sensor that translates the mechanical input forces received from the writing shaft into electrical forces for operation by electronics in the pen-stylus;

22

a writing shaft impact shock absorber that receives excess forces imparted to the pen-stylus to shield the force sensor from receiving the excess forces;

a rear erasure shaft, located at a rear end of the pen-stylus opposite to an end having the writing shaft, wherein the rear erasure shaft receives rear mechanical input forces as the user engages the rear end of the pen-stylus on the display of the tablet device;

a rear force sensor that translates the rear mechanical input forces received from the rear erasure shaft into electrical forces for operation by electronics in the pen-stylus; and

a rear shaft impact shock absorber that receives rear excess forces imparted to the pen-stylus to shield the rear force sensor from receiving the rear excess forces.

19. The pen-stylus of claim 18, wherein the rear excess forces arise from the pen-stylus being dropped onto a hard surface.

20. The pen-stylus of claim 18, further comprising:

an end erasure cap, wherein a body of the pen-stylus includes a rear eraser safety gap between the end erasure cap and a rear of the body of the pen-stylus, wherein when the body of the pen-stylus receives the rear excess forces along certain angles at the rear of the body of the pen-stylus, the end erasure safety cap absorbs the rear excess forces, shielding the pen-stylus from the rear excess forces.

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